

# Advanced Computer Networks – PCS223

## Assignment 8

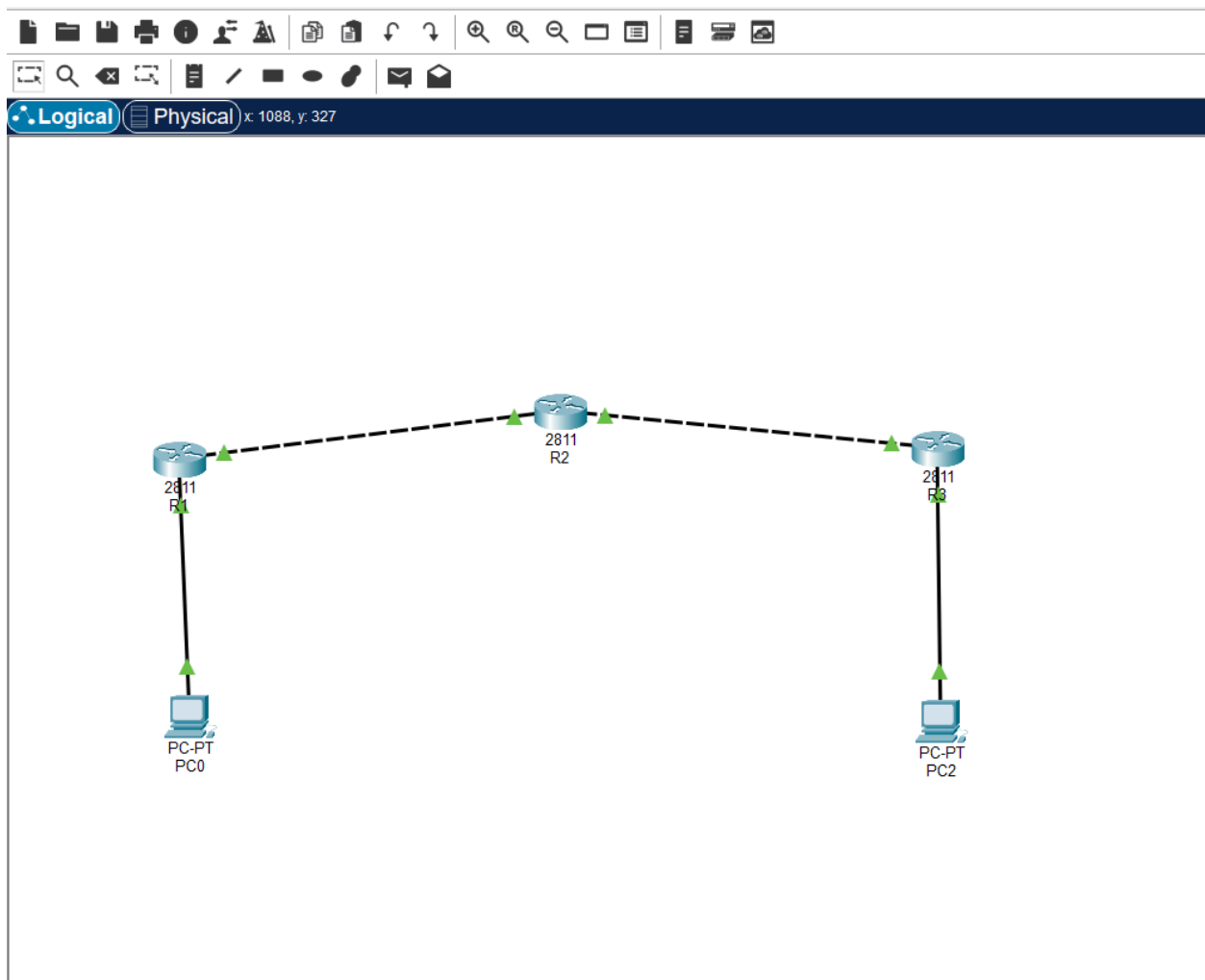
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### Q.1 Configuration and Analysis of BGP Routing

1. To understand the working of Border Gateway Protocol (BGP)
2. To configure External BGP (eBGP) between different Autonomous Systems
3. To analyze path selection using BGP attributes



Screenshot showing network topology and successful ping from PC2 to PC1

```
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up
20:03:36: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on Serial0/3/1 from LOADING to FULL, Loading Done
```

```
Router>enable
Router#show ip ospf neighbor
```

| Neighbor ID | Pri | State   | Dead Time | Address   | Interface   |
|-------------|-----|---------|-----------|-----------|-------------|
| 2.2.2.2     | 0   | FULL/ - | 00:00:39  | 100.1.1.1 | Serial0/3/0 |
| 3.3.3.3     | 0   | FULL/ - | 00:00:39  | 20.1.1.2  | Serial0/3/1 |

```
Router#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
    10.0.0.0/24 is subnetted, 1 subnets
O       10.2.2.0/24 [110/65] via 20.1.1.2, 00:08:04, Serial0/3/1
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.1.1.0/24 is directly connected, Serial0/3/1
L       20.1.1.1/32 is directly connected, Serial0/3/1
    100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       100.1.1.0/24 is directly connected, Serial0/3/0
L       100.1.1.2/32 is directly connected, Serial0/3/0
O       172.16.0.0/16 [110/65] via 100.1.1.1, 00:19:44, Serial0/3/0
```

```
Router#show ip ospf database
        OSPF Router with ID (1.1.1.1) (Process ID 1)
```

```
        Router Link States (Area 0)
```

| Link ID | ADV Router | Age  | Seq#       | Checksum | Link count |
|---------|------------|------|------------|----------|------------|
| 2.2.2.2 | 2.2.2.2    | 1184 | 0x80000003 | 0x00888c | 3          |
| 1.1.1.1 | 1.1.1.1    | 512  | 0x80000006 | 0x00d931 | 4          |
| 3.3.3.3 | 3.3.3.3    | 490  | 0x80000003 | 0x00afab | 3          |

```
Router#
```

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Press RETURN to get started.

Router>show ip ospf neighbor

| Neighbor ID | Pri | State   | Dead Time | Address   | Interface   |
|-------------|-----|---------|-----------|-----------|-------------|
| 1.1.1.1     | 0   | FULL/ - | 00:00:33  | 100.1.1.2 | Serial0/3/0 |

Router>show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP  
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area  
\* - candidate default, U - per-user static route, o - ODR  
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/24 is subnetted, 1 subnets  
O 10.2.2.0/24 [110/129] via 100.1.1.2, 00:11:53, Serial0/3/0  
20.0.0.0/24 is subnetted, 1 subnets  
O 20.1.1.0/24 [110/128] via 100.1.1.2, 00:14:16, Serial0/3/0  
100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks  
C 100.1.1.0/24 is directly connected, Serial0/3/0  
L 100.1.1.1/32 is directly connected, Serial0/3/0  
C 172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks  
C 172.16.0.0/16 is directly connected, FastEthernet0/0  
L 172.16.2.1/32 is directly connected, FastEthernet0/0

Router>show ip database

^  
% Invalid input detected at '^' marker.

Router>show ip ospf database

OSPF Router with ID (2.2.2.2) (Process ID 1)

Router Link States (Area 0)

| Link ID | ADV Router | Age  | Seq#       | Checksum | Link count |
|---------|------------|------|------------|----------|------------|
| 2.2.2.2 | 2.2.2.2    | 1415 | 0x80000003 | 0x00888c | 3          |
| 1.1.1.1 | 1.1.1.1    | 743  | 0x80000006 | 0x00d931 | 4          |
| 3.3.3.3 | 3.3.3.3    | 721  | 0x80000003 | 0x00afab | 3          |

Router>

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```
* incomplete command:
Router(config-router)#network 10.2.2.0 0.0.0.255 area 0
Router(config-router)#show ip ospf neighbour
      ^
% Invalid input detected at '^' marker.

Router(config-router)#exi
Router(config)#show ip ospf neighbour
      ^
% Invalid input detected at '^' marker.

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip ospf neighbour
      ^
% Invalid input detected at '^' marker.

Router#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
1.1.1.1          0    FULL/ -         00:00:37    20.1.1.1       Serial0/3/0
Router#show ip ospf database
        OSPF Router with ID (3.3.3.3) (Process ID 1)

        Router Link States (Area 0)

Link ID          ADV Router      Age          Seq#           Checksum Link count
2.2.2.2          2.2.2.2        827          0x80000003    0x00888c 3
1.1.1.1          1.1.1.1        155          0x80000006    0x00d931 4
3.3.3.3          3.3.3.3        132          0x80000003    0x00afab 3
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.2.2.0/24 is directly connected, FastEthernet0/0
L       10.2.2.1/32 is directly connected, FastEthernet0/0
C       20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.1.1.0/24 is directly connected, Serial0/3/0
L       20.1.1.2/32 is directly connected, Serial0/3/0
C       100.0.0.0/24 is subnetted, 1 subnets
O       100.1.1.0/24 [110/128] via 20.1.1.1, 00:02:47, Serial0/3/0
O       172.16.0.0/16 [110/129] via 20.1.1.1, 00:02:47, Serial0/3/0

Router#
```

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